# ANALYSIS OF STATISTICAL DISTRIBUTION CHANGES OF INPUT FEATURES IN NETWORK TRAFFIC CLASSIFICATION DOMAIN

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# MOTIVATION

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I want automatic detection/classification of ….

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3. Evaluate the model -> write the paper -> publish it -> go away from it

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**I have done all of these.**
**Is it done?**

**Examples:**

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- Detection of botnet --> run botnet in my network?!
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**Drift** is a term used in machine learning to describe how the performance of a machine learning model in production gets worse over time.

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- Will network traffic of botnet / TLS service change in time?
  --> Yes, it will. (updates of code, new versions, ....)
- Will it be a problem?
  --> Unfortunately yes, it will. (obsolescence of models leading to accuracy drop)

# ANALYSIS OF DATA DISTRIBUTION

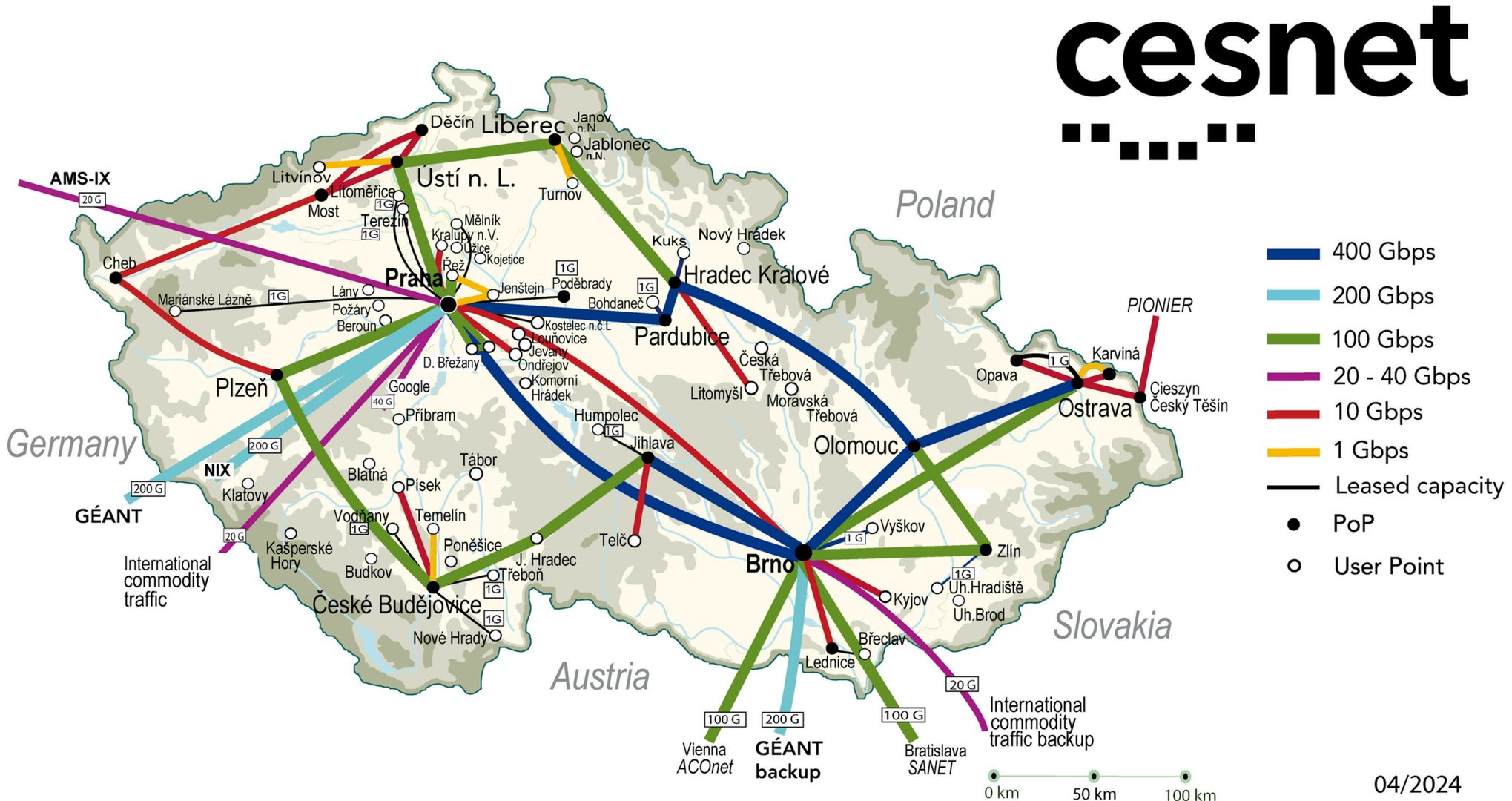
In the network traffic classification domain models rapidly lose accuracy and dataset relevance in time. Thus, we try to find out if concept drifts exist in the network traffic domain and also find out the type of concept drifts. Moreover, we try to find out if concept drifts repeat in time and how often.

# DATASET

- **CESNET-TLS-Year22** - one-year-long dataset of TLS network traffic from CESNET3
  - For work with the dataset, we used our open-source tool **CESNET-DataZoo**
  - 507,739,073 IP flows
  - 180 classes
  - 54 features



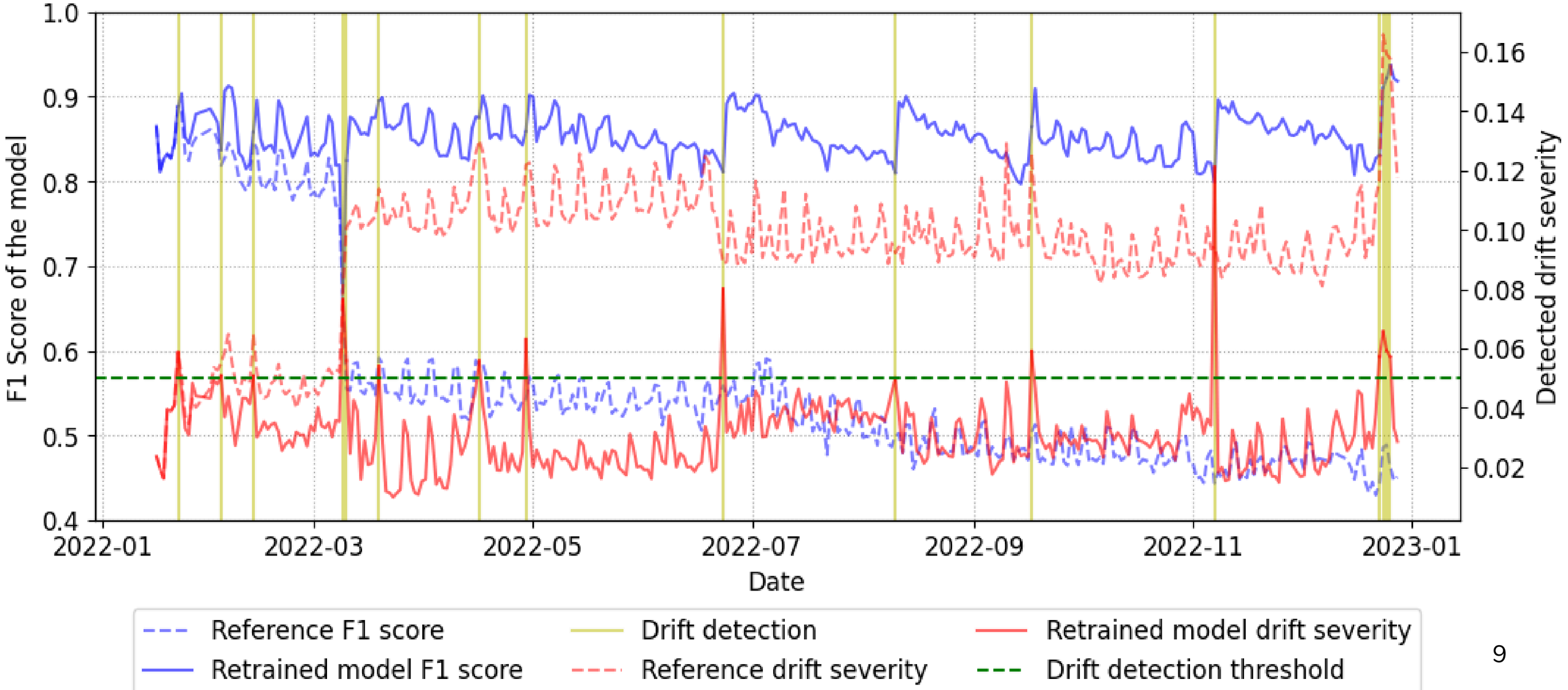
# CESNET3 NETWORK



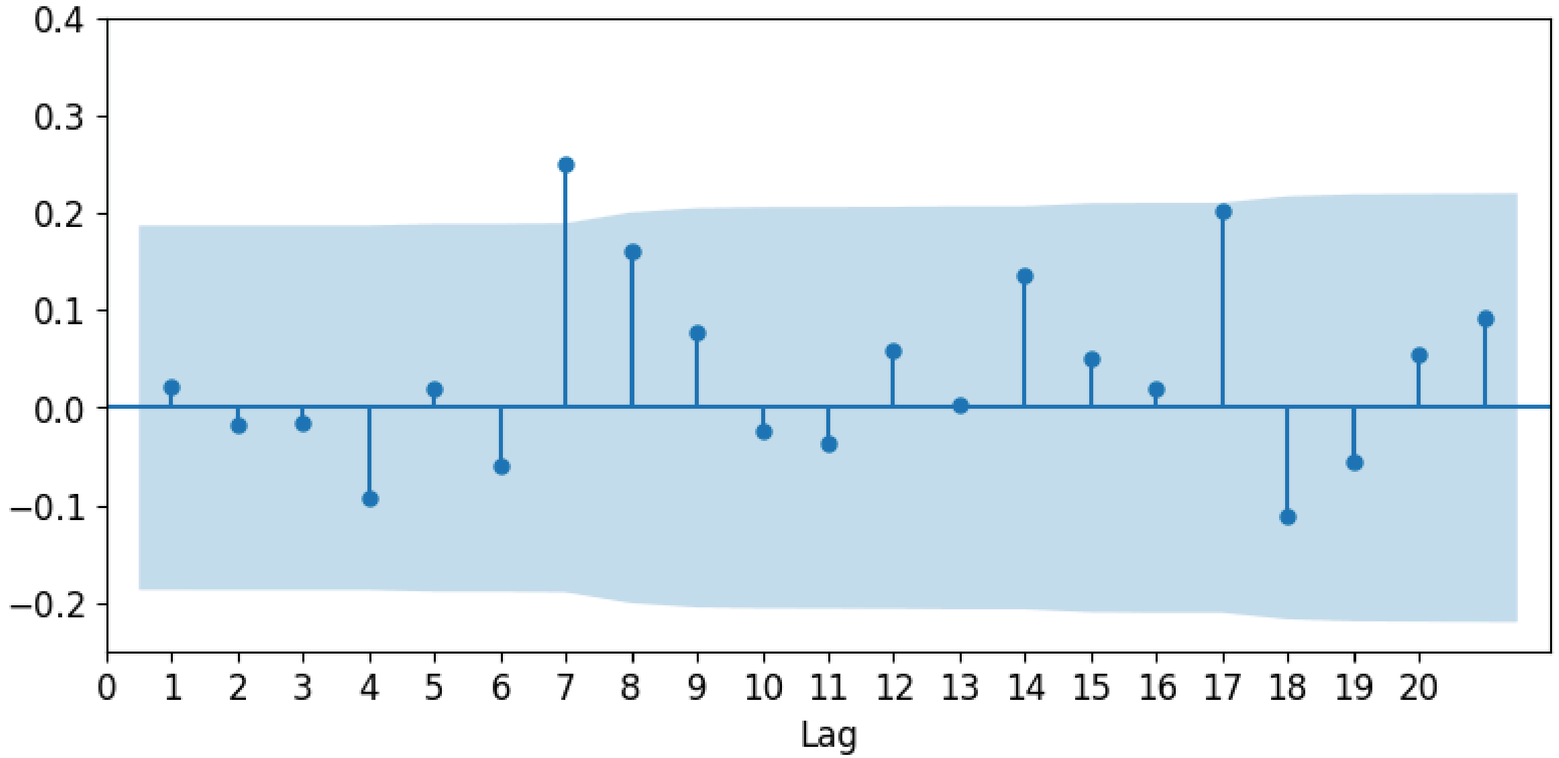



04/2024

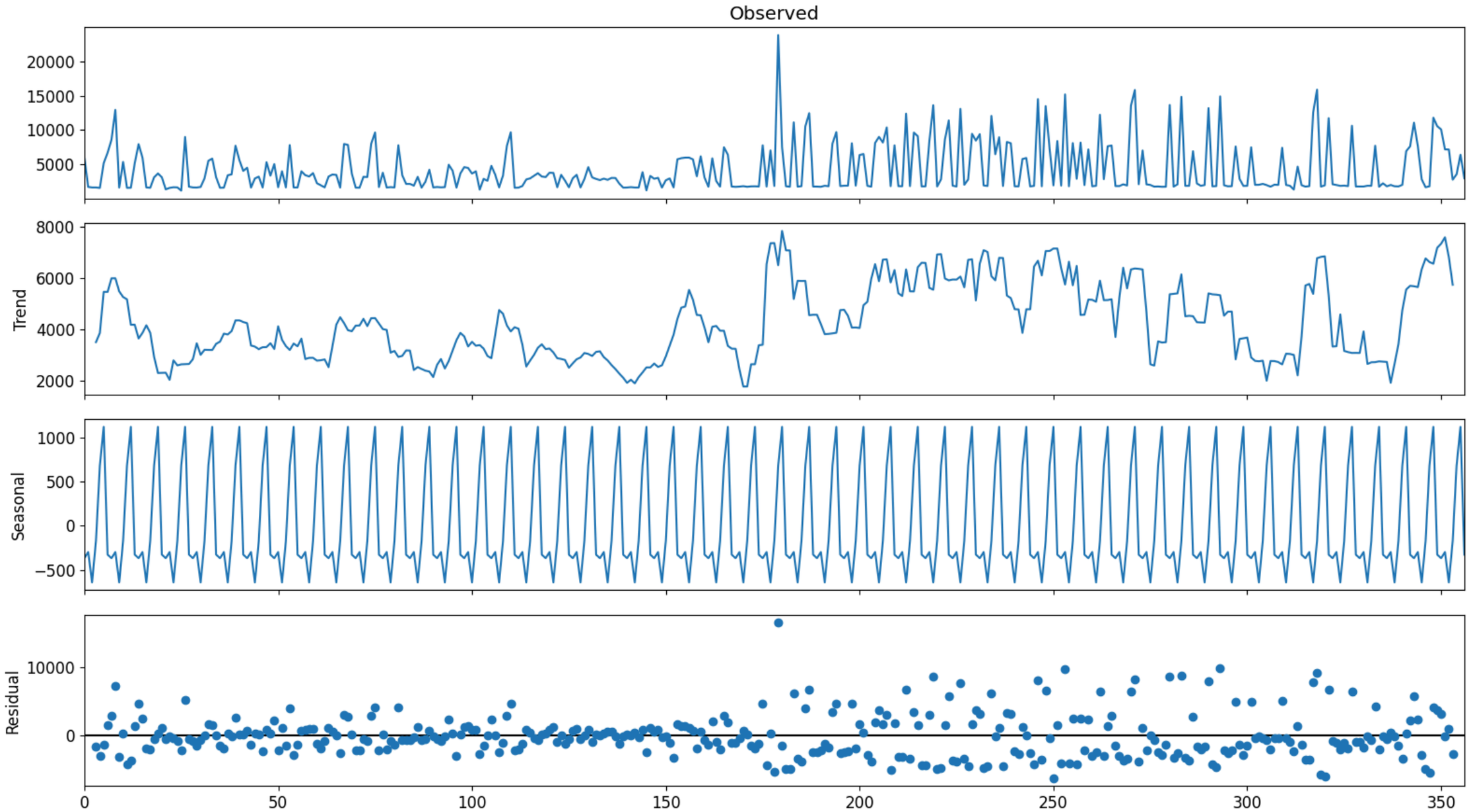
# DRIFT EVALUATION



# AUTOCORRELATION



# SEASONAL DECOMPOSE



# KOLMOGOROV–SMIRNOV TEST

The Kolmogorov-Smirnov (KS) test is used to test the goodness of fit of a given set of data to a theoretical distribution.

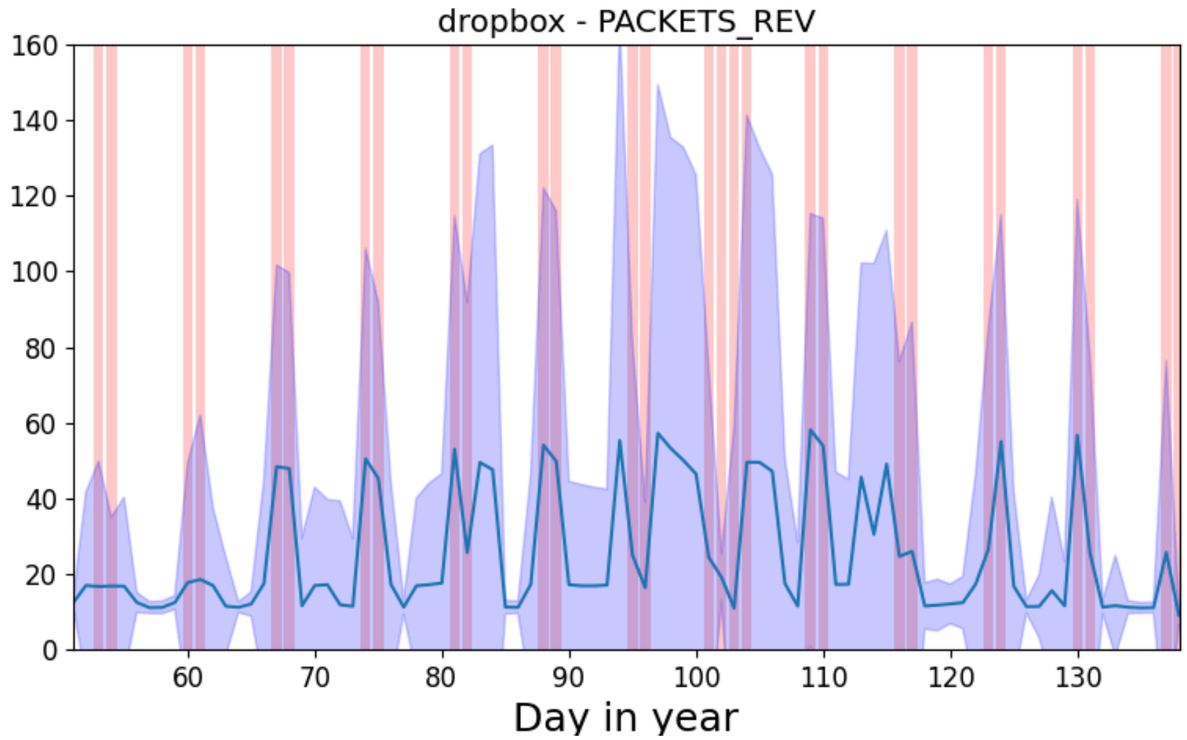
- We use KS test to compare the distribution of two set of data:
  - Weekdays data
  - Weekend data
- The KS test requires the data generated from one process
  ==> each class separately
- The KS test can be used only for one-dimensional data
  ==> each feature separately

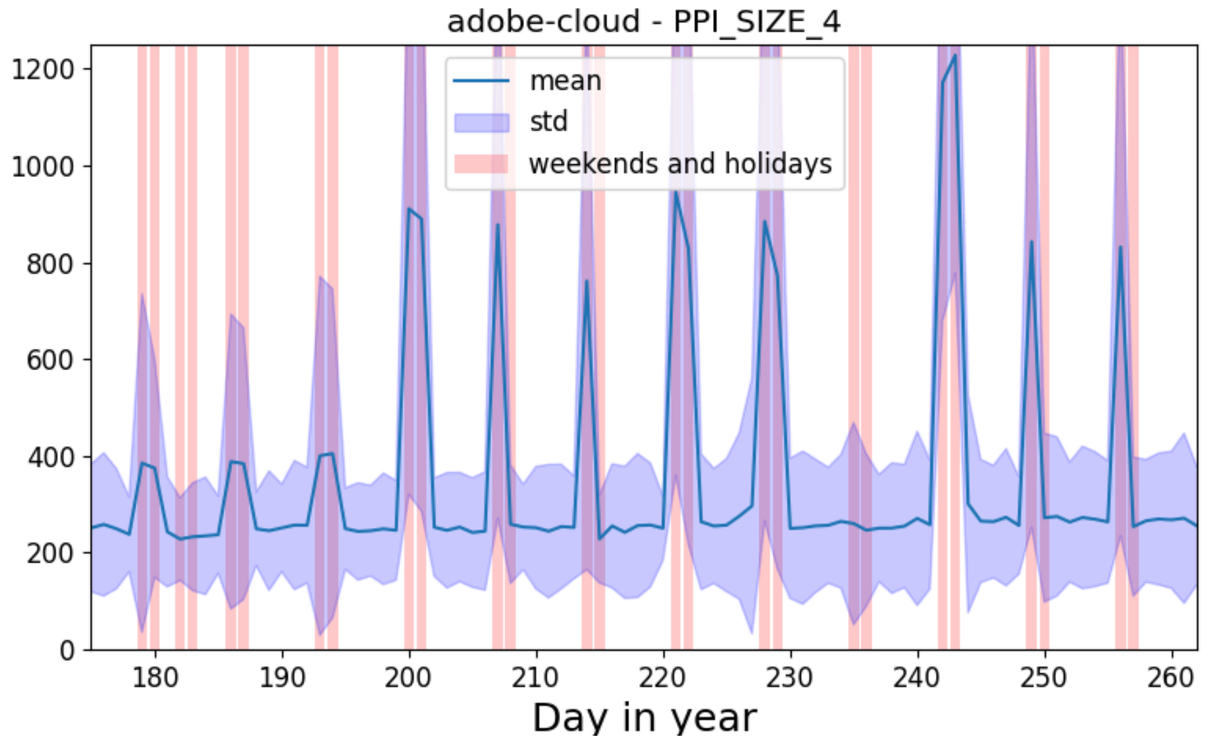
# KOLMOGOROV–SMIRNOV TEST

The experiment results shows:

- 38% of all tests (features X classes) result in the rejection of the null hypothesis ==> different distribution in Weekdays and Weekend
- Some features have usually the same distribution
  - TCP flags, PPI Inter Packet Times, Flow End Reason Features, …
- Some features have usually different distributions
  - Number of Bytes and Packets, Durations, PPI Sizes, Directions,…
- Some classes have the mostly same distribution
  - docker-registry, kaspersky, doh, chrome-remotedesktop, apple-updates, …
- Some classes have different distributions for most of the features
  - avast, spotify, slack, dropbox, snapchat, google-drive, youtube, …

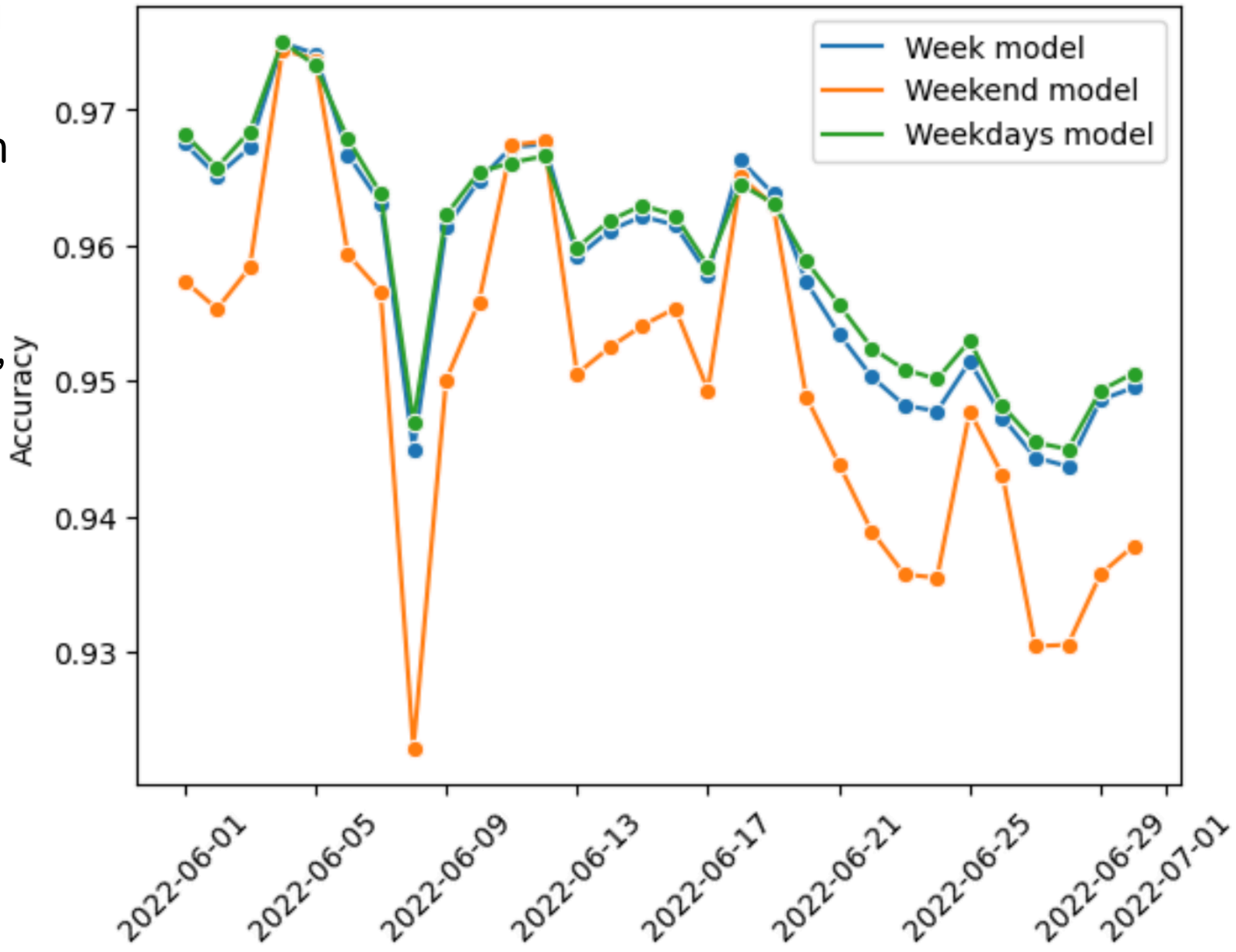
# WEEKEND PHENOMENON



# HOW DOES AFFECTS MODEL?

- Trained on May 2022 using CESNET-DataZoo
- Neural Network model from CESNET-Models
- The weekend phenomenon affects all models, however, despite the fact that the weekend model does not work on weekdays, the weekday model performs comparatively well on the weekend



# FUTURE WORK

- Multivariate distribution tests (like MMD)
- How much will distribution changes affect the model?
- How much will distribution change of one single class aggeft the model?
- On the CESNET-QUIC dataset, we also noticed that the traffic for each class differs between day hours and night hours. Exists also some Day/Night phenomenon?

## CESNET-DataZoo:



## CESNET-MODELS:

### CONTACTS

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